**Table 21.3** Minerals - Dietary sources, Functions and Deficiency disorders

| Minerals | Sources | Functions | Deficiency disorders |
|---|---|---|---|
| **Macronutrients** | | | |
| **Calcium** | Dairy products, beans, cabbage, eggs, fish | Constituent of bones and enamel of teeth, clotting of blood and controls muscle contraction. | Bone deformities, poor skeletal growth, osteoporosis in adults. |
| **Sodium** | Common salt | Maintains fluid balance and involved in neurotransmission. | Muscular cramps, nerve impulses do not get transmitted. |
| **Potassium** | Banana, sweet potato, nuts, whole grains, citrus fruits | Regulates nerve and muscle activity. | Muscular fatigue, nerve impulses do not get transmitted. |
| **Micronutrients** | | | |
| **Iron** | Spinach, dates, greens, broccoli, whole cereals, nuts, fish, liver | Important component of haemoglobin. | Anaemia |
| **Iodine** | Milk, Seafood, Iodised salt | Formation of thyroid hormones. | Goitre |

**Kwashiorkar:** It is a condition of severe protein deficiency. It affects children between 1-5 years of age, whose diet mainly consists of carbohydrates but lack in proteins.

**Marasmus:** It usually affects infants below the age of one year when the diet is poor in carbohydrates, fats and proteins.



**Kwashiorkar**



**Marasmus**

**Figure 21.1** Malnutrition

## 21.3 Food Hygiene

Food spoilage is an undesirable change in the normal state of food and is not suitable for human consumption. Poor personal hygiene may allow pathogenic microorganisms to cause food spoilage. Signs of food spoilage include a changes in appearance, colour, texture, odour and taste. Factors responsible for Food Spoilage are given below.

**Internal factors:** It includes enzymatic activities and moisture content of the food.

**External factors:** It includes adulterants in food, contaminated utensils and equipment, unhygienic cooking area and lack of storage facilities.

## 21.4 Food Preservation

Food preservation is the process of prevention of food from decay or spoilage, by storing in a condition fit for future use. Food is preserved to:

- increase the shelf life of food
- retain the colour, texture, flavour and nutritive value
- increase food supply
- decrease wastage of food

### 21.4.1 Methods of Food Preservation

The various method of food preservation are explained below.



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